

Enthusiast COURSE

PHASE : (RBT)

TARGET : PRE-MEDICAL 2025

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 28-03-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	3	3	2	4	4	3	2	2	3	2	1	2	3	2	2	3	2	4	3	1	2	2	4	1	3	2	4	3	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	3	3	2	3	1	4	2	2	1	2	2	1	2	2	1	1	4	3	1	4	2	1	4	1	2	3	4	4	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	1	1	2	1	3	2	3	4	3	1	2	1	4	3	3	1	1	1	1	3	3	1	2	2	4	1	2	2	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	1	3	2	3	1	4	2	4	3	4	1	2	2	1	2	1	1	4	3	3	3	4	2	3	4	2	4	2	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	3	1	1	1	4	1	2	3	3	2	3	2	4	4	3	3	1	2	1	4	2	4	3	2	4	2	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	3	4	4	4	2	2	4	2	2	2	1	1	1	4	3	1	2	3	4	4	3	2	3	2	1	2	1	2	2

HINT - SHEET

1. Ans (3)

$$\begin{aligned}\because \cos 2\theta &= 2\cos^2\theta - 1 \\ \cos^2\theta &= \frac{1 + \cos 2\theta}{2} \\ \therefore \cos^2 20^\circ &= \frac{1 + \cos 40^\circ}{2}\end{aligned}$$

2. Ans (3)

3. Ans (3)

Explain - Find equation of line from given graph.

Concept - First find slope of line then apply Trigonometry and find constant c.

Formula - Slope = $\tan \theta = m$, $y = mx + c$

Calculation-

$$\begin{aligned}\text{Slope} = m &= \tan 45^\circ \\ &= 1\end{aligned}$$

intercept of line of y-axis is

$$c = 3$$

$$\text{So } y = mx + c$$

$$y = x + 3$$

4. Ans (2)

$$\begin{aligned}\int_0^{\pi/4} \frac{\sin 2x}{2} dx &= \frac{1}{2} \cdot \frac{1}{2} [-\cos 2x]_0^{\pi/4} \\ &= -\frac{1}{4} \left[\cos \frac{\pi}{2} - \cos 0 \right] = \frac{1}{4}\end{aligned}$$

5. Ans (4)

$$\begin{aligned}y_{\text{avg.}} &= \langle y \rangle = \frac{\int_0^{20} y \cdot dx}{\int_0^{20} dx} \\ &= \frac{\left(\frac{1}{2} \times 10 \times 25 + 10 \times 25 \right)}{20} \\ &= \frac{375}{20} = 18.75\end{aligned}$$

6. Ans (4)

$$\begin{aligned}\text{Let } \vec{c} &= 3\lambda \hat{i} + 4\lambda \hat{j} \\ \text{Now, } (3\lambda)^2 + (4\lambda)^2 &= 7^2 + 24^2 \\ \Rightarrow \lambda &= 5 \\ \therefore \vec{c} &= 15\hat{i} + 20\hat{j}\end{aligned}$$

7. **Ans (3)**

$$\sqrt{(0.3)^2 + b^2 + (0.4)^2} = 1$$

$$0.09 + b^2 + 0.16 = 1$$

$$b = \sqrt{0.75}$$

8. **Ans (2)**

$$\vec{A} \cdot \vec{B} = AB \cos \theta \text{ (scalar)}$$

9. **Ans (2)**

$$A_x = A \cos 45^\circ = 5\sqrt{2} \times \frac{1}{\sqrt{2}} = 5$$

$$A_y = A \sin 45^\circ = 5\sqrt{2} \times \frac{1}{\sqrt{2}} = 5$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

$$\vec{A} = 5\hat{i} + 5\hat{j}$$

10. **Ans (3)**

11. **Ans (2)**

$$P = \frac{a - x^2}{bt}$$

By homogeneity $\Rightarrow [a] = [x^2] = L^2$

$$\text{So, } [P] = \frac{L^2}{[b]T} \Rightarrow ML^2T^{-3} = \frac{L^2}{[b]T}$$

$$[b] = \frac{L^2}{ML^2T^{-2}}$$

$$\Rightarrow M^{-1}L^0T^2$$

12. **Ans (1)**

As a is a dimensionless $a = \frac{1}{t} = [T^{-1}]$ further, as x and $\frac{v_0}{a}$ have same dimensions.

$$L = \frac{v_0}{T^{-1}}$$

$$v_0 = [M^0L^1T^{-1}]$$

13. **Ans (2)**

Torque = $F \times r_{\perp}$ Nm

Stress = $\frac{\text{Force}}{\text{Area}}$ N/m²

Latent heat = $\frac{\text{Energy}}{\text{Mass}}$ J Kg⁻¹

Power = $\frac{\text{Work}}{\text{Time}}$ N ms⁻¹

A-III, B-IV, C-II, D-I

14. **Ans (3)**

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Pressure gradient = $\frac{dp}{dx} = \frac{[ML^{-1}T^{-2}]}{[L]}$

$$= [M^1L^{-2}T^{-2}]$$

Energy density = $\frac{\text{energy}}{\text{volume}} = \frac{[ML^2T^{-2}]}{[L^3]}$

$$= [M^1L^{-1}T^{-2}]$$

Electric field = $\frac{\text{Force}}{\text{charge}} = \frac{[MLT^{-2}]}{[A \cdot T]}$

$$= [M^1L^1T^{-3}A^{-1}]$$

Latent heat = $\frac{\text{heat}}{\text{mass}} = \frac{[ML^2T^{-2}]}{[M]}$

$$= [M^0L^2T^{-2}]$$

15. **Ans (2)**

$$1\text{MSD} = \frac{1}{10} = 0.1 \text{ cm}$$

$$10\text{VSD} = 8\text{MSD}$$

$$1\text{VSD} = \frac{8}{10} \text{ MSD}$$

$$\text{L.C.} = 1\text{MSD} - 1\text{VSD}$$

$$= 1\text{MSD} - \frac{8}{10} \text{ MSD} = \frac{2}{10} \text{ MSD}$$

$$= \frac{2}{10} \times 0.1 = 0.02 \text{ cm}$$

16. **Ans (2)**

Thickness of body = $11\text{mm} + 65 \left(\frac{1\text{mm}}{100} \right)$

$$= 11.65$$

17. **Ans (3)**

As $P = \frac{v^2}{R}$

$$\frac{\Delta P}{P} = \frac{2\Delta v}{v} \quad (R \rightarrow \text{constant})$$

$$\frac{\Delta P}{P} \times 100 = 2(2.5\%) = 5\%$$

18. **Ans (2)**

Ques explanation: To check the impact of velocity components on minimum time in crossing the river.

Solution: The assertion is correct: to cross the river in minimum time, the swimmer should swim perpendicular to the current.

The reason is correct because the river flow does not assist in crossing the river; it only affects the swimmer's horizontal displacement.

Concept: River-Boat Concept

Formula: Not required

Question level: Easy

19. Ans (4)

$$V_{avg} = \frac{x}{\frac{2x}{V_1} + \frac{3x}{V_2}}$$

$$V_{avg} = \frac{5}{\frac{2}{V_1} + \frac{3}{V_2}} = \frac{5V_1V_2}{3V_1 + 2V_2}$$

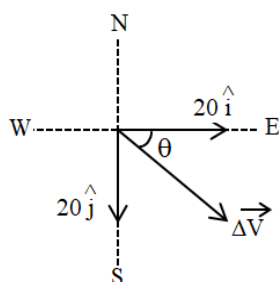
20. Ans (3)

$$\vec{v} = \vec{u} + \vec{a}t = (3\hat{i} + 4\hat{j}) + (0.4\hat{i} + 0.3\hat{j})(10)$$

$$= 7\hat{i} + 7\hat{j}$$

$$\text{So speed} = |\vec{v}| = 7\sqrt{2} \text{ units}$$

22. Ans (2)



$$\vec{v}_i = 20\hat{j} \text{ m/s}, \vec{v}_f = 20\hat{i} \text{ m/s}$$

$$\Delta\vec{v} = \vec{v}_f - \vec{v}_i = (20\hat{i} - 20\hat{j}) \text{ m/s}$$

$$|\Delta\vec{v}| = 20\sqrt{2} \text{ m/s}$$

$$\tan \theta = \frac{20}{20} = 1$$

$$\theta = 45^\circ$$

south-east direction.

23. Ans (2)

$$f = at$$

$$\frac{dv}{dt} = at$$

$$\int_u^v dv = \int_0^t at dt$$

$$v - u = \frac{at^2}{2} \Rightarrow v = \frac{u + at^2}{2}$$

24. Ans (4)

$$v = \frac{ds}{dt} = \beta + 2\gamma t + 4\delta t^3 = \beta \text{ (at } t = 0)$$

$$a = \frac{dv}{dt} = 2\gamma + 12\delta t^2 = 2\gamma \text{ (at } t = 0)$$

$$\text{Ratio} = \frac{\beta}{2\gamma}$$

25. Ans (1)

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B. Given Data:

Projectile motion: Motion of an object thrown with an initial velocity under the

- influence of gravity.

C. Concept:

- **Projectile Motion:**

Characterized by two independent

- motions:

- Uniform horizontal motion

Uniformly accelerated vertical

- motion (due to gravity)

Acceleration due to Gravity (g): A constant downward acceleration acting on all objects

- near the Earth's surface.

D. Mathematical Calculation:

- **Vertical Motion:**

Can be described using the equation:

- $v = u - gt$

- where:

- v: final vertical velocity

u: initial vertical

- velocity

g: acceleration due to

- gravity

- t: time

As time (t) increases during the

upward motion, the term 'gt'

increases, leading to a decrease in the

- final vertical velocity (v).

E. Final Answer:

Both Assertion & Reason are true and the

Reason is the correct explanation of the

- Assertion. option 1

26. Ans (3)

$$S = \frac{u^2}{2a}$$

If u becomes 2 times, stopping distance becomes 4 times i.e., 8m.

27. Ans (2)

$$\begin{aligned} \because S_{nth} &= u + \frac{a}{2}(2x-1) = 0 + \frac{g}{2}(2x-1) \\ \therefore S_{nth} &\propto (2x-1) \\ \therefore S_{1st} : S_{3rd} : S_{6th} &= (2 \times 1 - 1) : (2 \times 3 - 1) : (2 \times 6 - 1) \\ &= 1 : 5 : 11 \end{aligned}$$

28. Ans (4)

$$\begin{aligned} t_{Net} &= t_{up} + t_{down} = \frac{u}{g} + \frac{v}{g} = \frac{u+v}{g} \\ t &= \frac{u + \sqrt{u^2 + 2gh}}{g} \end{aligned}$$

29. Ans (3)

30. Ans (3)

$$\begin{aligned} \because H &= \frac{u^2}{2g}, H \propto v^2 \\ \therefore \frac{H_1}{H_2} &= \left(\frac{u_1}{u_2}\right)^2, \frac{20}{H_2} = \left(\frac{u}{2u}\right)^2, H_2 = 80\text{m} \end{aligned}$$

31. Ans (2)

$$\begin{aligned} x_1 &= \frac{1}{2}at^2 \\ x_2 &= vt \\ x_1 - x_2 &= \frac{1}{2}at^2 - vt \\ Y &= k_1x^2 - k_2x \end{aligned}$$

32. Ans (3)

$$\begin{aligned} \text{velocity} \rightarrow \text{In AB} \rightarrow v &= \frac{20-0}{2-0} = 10\text{ m/s} \\ \text{In BC} \rightarrow v &= \frac{20-20}{4-2} = 0 \\ \text{In CD} \rightarrow v &= \frac{60-20}{6-4} = 20\text{ m/s} \\ \text{In DE} \rightarrow v &= \frac{60-40}{8-6} = 0 \end{aligned}$$

maximum velocity during CD

33. Ans (3)

Velocity is slope of x-t graph

$$\begin{aligned} V &= \frac{dx}{dt} = \tan \theta \\ \frac{V_1}{V_2} &= \frac{\tan \theta_1}{\tan \theta_2} = \frac{\tan 30^\circ}{\tan 45^\circ} = \frac{1}{\sqrt{3}} \end{aligned}$$

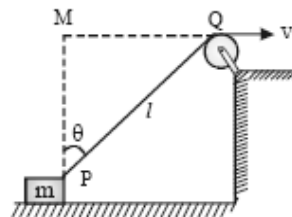
34. Ans (2)

$$x \propto t^2$$

35. Ans (3)

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Here θ and the length PQ vary with time. Let $QP = l$ at any instant.



$$PM = l \cos \theta$$

$$MQ = l \sin \theta$$

$$v = \frac{dl}{dt} \text{ (given)}$$

As the block does not move vertically.

$$\frac{d}{dt}(l \cos \theta) = 0$$

$$\text{Further, } \frac{d}{dt}(MQ) = \text{required velocity}$$

$$\text{or } \frac{d}{dt}(l \sin \theta) = v \text{ (say)}$$

Solving these equation, we get :

$$V = \frac{v}{\sin \theta}$$

36. Ans (1)

$$\theta = 45^\circ; R = \frac{u^2 \sin(2 \times 45^\circ)}{g} = \frac{u^2}{g}$$

37. Ans (4)

For same range $\theta_1 + \theta_2 = 90^\circ$

$$\theta_1 = \frac{\pi}{3} \text{ rad} = 60^\circ$$

$$\therefore \theta_2 = 90^\circ - \theta_1 = 30^\circ$$

$$y_1 = \frac{u^2 \sin^2 60^\circ}{2g} = \frac{u^2 \times 3}{2g \times 4} \Rightarrow \frac{u^2}{8g} = \frac{y_1}{3}$$

$$y_2 = \frac{u^2 \sin^2 30^\circ}{2g} = \frac{u^2 \times 1}{2g \times 4} = \frac{y_1}{3}$$

38. Ans (2)

$$\text{Time of flight, } T = \frac{2u_y}{g} = 2 \frac{20}{10} \times 2 = 4 \text{ sec}$$

39. Ans (2)

Asking about : T, R & H Via graph

Concept : Interpret different quantity by trajectory.

Solution/Explanation/Calculation :

$$R \frac{2u_x u_y}{g}, H = \frac{u_{xy}^2}{2g}, T = \frac{2u_y}{g}$$

as max. height is same for all hence u_y is same for all

But Range are diff. hence u_x is diff.

hence launch speed in diff. for all the three.

Hence, the correct answer is option (2).

40. Ans (1)

$$R = 2H \Rightarrow \tan \theta = \frac{4}{2} = 2$$

$$R = \frac{2V^2 \left(\frac{2}{\sqrt{5}} \right) \left(\frac{1}{\sqrt{5}} \right)}{g} = \frac{4V^2}{5g}$$

41. Ans (2)

$$R = \frac{u^2 \sin 2\theta}{g} \quad R \propto \sin 2\theta$$

$$\therefore \frac{R_1}{R_2} = \frac{\sin 2\theta_1}{\sin 2\theta_2}, \quad \frac{50}{R_2} = \frac{\sin 30^\circ}{\sin 90^\circ}$$

$$R_2 = 100 \text{ m}$$

42. Ans (2)

$$\frac{R}{H} = \frac{\frac{u^2 \times 2 \sin \theta \cos \theta}{g}}{\frac{u^2 \sin^2 \theta}{2g}} = 4 \cot \theta$$

43. Ans (1)

Given equation $y = \sqrt{3}x \left(1 - \frac{x}{5\sqrt{3}} \right)$

Equation trajectory $y = x \tan \theta \left(1 - \frac{x}{R} \right)$

By comparison

$$(1) \tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ = \frac{\pi}{3} \text{ rad}$$

$$(2) R = 5\sqrt{3}$$

$$(3) \tan \theta = \frac{4H}{R} \Rightarrow H = \frac{R}{4} \tan \theta = \frac{5\sqrt{3}}{4} \times \sqrt{3} = \frac{15}{4}$$

$$(4) H_{\max} = \frac{u^2 \sin^2 60^\circ}{2g} \Rightarrow \frac{15}{4} = \frac{u^2}{20} \cdot \frac{3}{4} \Rightarrow u = 10 \text{ m/s}$$

44. Ans (2)

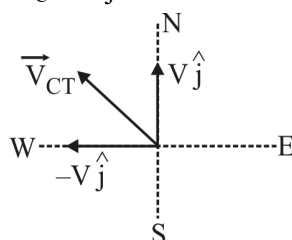
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$$\begin{aligned} \frac{120}{t} &= u \cos \theta & (u \sin \theta)2 - \frac{1}{2}g(2)^2 \\ \frac{120}{4} &= u \cos \theta & = (u \sin \theta)6 - \frac{1}{2}g(6)^2 \\ 30 \text{ m/s} &= u \cos \theta = u_x & \frac{1}{2}g[6^2 - 2^2] = (u \sin \theta) \\ & & \frac{1}{2}10 \left[\frac{32}{4} \right] = u \sin \theta \\ & & 40 \text{ m/s} = u \sin \theta \\ R &= \frac{2u^2 \sin \theta \cos \theta}{g} = \frac{2 \times 30 \times 40}{10} = 240 \text{ m} \end{aligned}$$

45. Ans (2)

$$\vec{V}_T = V \hat{i}$$

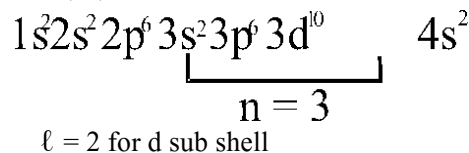
$$\vec{V}_C = V \hat{j}$$



$$\vec{V}_{CT} = \vec{V}_C - \vec{V}_T = V \hat{j} - V \hat{i}$$

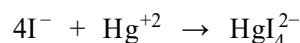
direction of $\vec{V}_{CT}$:- West-North

47. Ans (1)



57. Ans (3)

I.R.



$$t = 0 \quad 1 \text{ mol} \quad 1 \text{ mol}$$

$$t = \infty \quad 0 \quad 0.75 \text{ mol} \quad 0.25 \text{ mol}$$

60. Ans (1)

$$\begin{aligned} \text{mass} &= 2.5 \times 40 \\ &= 100 \text{ g} \end{aligned}$$

68. Ans (3)

NCERT-XI (Part-1) Pg. # 91

69. Ans (4)

1 mole = 2204 L is only valid for gases at STP

72. Ans (2)

VWR > SBCR

VWR $\approx 2 \times$ CR

75. **Ans (3)**
Based classification of group. in periodic table
77. **Ans (1)**
NCERT problem 3.31 page 98
78. **Ans (1)**
Refer - NCERT D & F block chapter
80. **Ans (1)**
NCERT-XI, Part-I Pg.-94
81. **Ans (3)**
Problem 3.5 NCERT
83. **Ans (1)**
Theory based NCERT Page No.75
91. **Ans (3)**
NCERT Pg. # 209
92. **Ans (1)**
NCERT-XII, Pg. # 209
94. **Ans (2)**
NCERT-XII, Pg. # 212
95. **Ans (3)**
NCERT-XII-Page No. # 211 (E), 231, 232 (H)
96. **Ans (1)**
NCERT Pg#212
97. **Ans (4)**
NCERT Pg # 212
98. **Ans (2)**
NCERT Pg. No. 180
99. **Ans (4)**
NCERT-XII, Pg.# 153
100. **Ans (3)**
NCERT Pg # 182/(H)199
103. **Ans (2)**
NCERT Pg # 213
104. **Ans (2)**
NCERT, Pg. # 185
105. **Ans (1)**
NCERT XII, Pg # 179

108. **Ans (1)** Join Telegram: @NEETxNOAH
NCERT Pg. # 210
110. **Ans (3)**
NCERT-XII, Pg # 198, 200
111. **Ans (3)**
NCERT XII, Page # 178
112. **Ans (3)**
NCERT XII, Pg. No. 208
113. **Ans (4)**
NCERT XII Pg # 202
114. **Ans (2)**
NCERT Page No. # 208 / 228
115. **Ans (3)**
NCERT XII, Pg # 182
116. **Ans (4)**
NCERT XII Pg # 178
117. **Ans (2)**
The correct answer is:
(2) Modified bacterial plasmid.
Explanation: pBR322 is a modified bacterial plasmid that is commonly used as a vector for cloning genes in E. coli. It was engineered from a natural plasmid to include specific features like antibiotic resistance markers, making it easier to select transformed cells.
120. **Ans (1)**
NCERT, Pg # 204
122. **Ans (2)**
NCERT XII Pg # 208, 209, 212
123. **Ans (3)**
NCERT-XII, Pg # 196 & 197
124. **Ans (1)**
NCERT-XII, Pg # 212, 213
127. **Ans (4)**
BamHI is a restriction enzyme that cuts at a specific sequence, commonly used in cloning and plasmid manipulation.
Sal I is another restriction enzyme used to cut at a specific site, often in recombinant DNA technology.

128. Ans (1)

Problem statement : Asking About :

Match column of gene gun, gene therapy, gene closing, genome.

Concept :

Gene gun - Vector less gene transfer.

Gene therapy - Adenosine deaminase deficiency.

Solution / Explanation :

Gene gun used for gene transfer.

Gene therapy replacement of a faulty gene by normal gene.

Gene closing to obtain identical copies of a gene.

Genome is DNA of organism's cell.

129. Ans (2)

NCERT-XII, Pg # 187

130. Ans (3)

NCERT-XII, Pg. # 151

131. Ans (3)

NCERT (XII) Pg. # 153

133. Ans (3)

NCERT XII, Pg. # 183

134. Ans (2)

NCERT-XII, Pg. # 166, 171

136. Ans (4)

The correct answer is: 4. Yeast.

Explanation: Yeast is also involved in the fermentation process for idli and dosa. The yeast ferments the carbohydrates in the dough, producing carbon dioxide, which helps the dough rise and become fluffy.

137. Ans (3)

NCERT XII, Pg. # 183

139. Ans (1)

EcoRI is the restriction enzyme that creates sticky ends on the vector and foreign DNA. **DNA ligase** is used to join the sticky ends and form recombinant DNA.

140. Ans (2)

NCERT Pg. # 152

141. Ans (1)

NCERT-XII, Page # 184 (10.3)

145. Ans (3)

The correct answer is 3: **Saccharomyces cerevisiae**.

Explanation: The organism also called **brewer's yeast** is **Saccharomyces cerevisiae**. It is used in the brewing industry for fermenting beer and in baking industry to help dough rise.

146. Ans (2)

The correct answer is: 2. **Micropropagation**

Explanation: **Micropropagation** is the method of producing thousands of plants through tissue culture. It involves the invitro cultivation of plant cells, tissues or organs to regenerate whole plants, which can then be propagated on a large scale. This technique is widely used for the rapid multiplication of plants with desirable traits.

147. Ans (4)

While curd (yogurt) has various nutritional benefits over milk, it does not provide additional proteins compared to milk. The protein content in curd is similar to that in milk; however, the proteins in curd are often more easily digestible due to the fermentation process.

148. Ans (2)

Propionibacterium shermanii is the bacterium responsible for the characteristic large holes in Swiss cheese (Emmental). During the fermentation process, this bacterium produces carbon dioxide (CO₂) as a byproduct of fermentation, which forms the holes or "eyes" in the cheese.

149. Ans (3)

Wine is made through the process of fermentation of grapes or other fruits, not distillation.

Fermentation converts sugars into alcohol, and the product is wine.

151. Ans (2)

Staphylococcus bacteria were crucial in the discovery of penicillin. In 1928, Alexander Fleming discovered penicillin when he noticed that a mold (*Penicillium notatum*) had killed the bacteria *Staphylococcus aureus* in a petri dish. This led to the development of penicillin as the first widely used antibiotic.

Transgenic animals, such as goats and sheep, have been genetically engineered to produce human AAT in their milk. This AAT can be purified and used to treat individuals with AAT deficiency, helping to prevent or slow the progression of emphysema.

152. Ans (3)

A.	Citric acid	ii.	<i>Aspergillus niger</i>
B.	Streptokinase	i.	Haemolytic streptococcus
C.	Cyclosporin	iv.	<i>Trichoderma polysporum</i>
D.	Statins	iii.	<i>Monascus purpureus</i>

153. Ans (4)

IARI (Indian Agricultural Research Institute) and KVIC (Khadi and Village Industries Commission) both played significant roles in the development and promotion of biogas technology in India.

- IARI contributed through research on the agricultural aspects of biogas production, such as optimizing the use of organic waste for biogas generation.
- KVIC was instrumental in promoting the practical application of biogas plants, especially in rural areas, and supported their widespread adoption for energy production, particularly in the 1970s.

154. Ans (4)

Cyanobacteria, particularly nitrogen-fixing cyanobacteria, are commonly used in rice cultivation. They help improve soil fertility by fixing atmospheric nitrogen, which is beneficial for the rice plants. These bacteria are often applied to rice paddies as a natural biofertilizer, particularly in flooded conditions where they thrive.

158. Ans (4)

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Baculoviruses are a group of viruses used as biological control agents in pest management. Their use is particularly desirable in several conditions:

1. When they are used as part of Integrated Pest Management (IPM): Baculoviruses can be an effective part of IPM, where they help control pest populations without harming the environment or non-target organisms.
2. When an ecologically sensitive area is being treated: Baculoviruses are specific to certain insect pests and do not harm other wildlife, making them ideal for use in ecologically sensitive areas.
3. When beneficial insects are being conserved: Since baculoviruses are highly specific to certain pests, they do not negatively affect beneficial insects (such as pollinators or natural predators), making them safer to use in conservation efforts.

159. Ans (2)

Viruses used as bioinsecticides are host-specific, meaning they only target specific insect species. This is a significant advantage because it reduces the impact on non-target organisms, making them safer for the environment compared to broad-spectrum chemical insecticides.

160. Ans (2)

NCERT XII Pg. # 180, Fig.10.2

161. Ans (2)

Nucleopolyhedrosis viruses (NPVs) are species-specific and narrow-spectrum biological agents used in insect pest control. They specifically target certain insect species, typically those in the order Lepidoptera (e.g., caterpillars), and cause disease by infecting the insect's cells, eventually leading to the insect's death.

162. Ans (1)

Bacillus thuringiensis (Bt) produces toxins that specifically affect the gut of insects. When an insect consumes Bt-contaminated material, the toxins in the bacteria disrupt the gut lining, leading to paralysis, starvation, and death of the insect. The toxins primarily target the insect's digestive system, causing it to stop feeding.

163. Ans (1)

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164. Ans (1)

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167. Ans (1)

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178. Ans (1)

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